

June 2002 Ext 1
(HY)

3U ASS
2002 PLC

-3-

Question 1 (11 marks) Start a new page.

Marks:

(a) Find $\frac{d}{dx} \tan^{-1} \frac{x}{2}$ 1

(b) Find $\int \frac{1}{\sqrt{2-x^2}} dx$ 1

- (c) i) Sketch the graph of $y = 2 \cos^{-1} \left(\frac{x}{3} \right)$
ii) State the domain and range of this function. 3

- (d) Without the aid of calculus, sketch the graph of the following polynomial function, clearly indicating all intercepts with the axes.

$$f(x) = -x(x+2)^2(x-1) \quad 2$$

- (e) Use the substitution $u = x^2 - 1$ to evaluate:

$$\int_1^2 2x(x^2 - 1)^4 dx \quad 4$$

Question 2 (11 marks) Start a new page.

Marks:

- (a) A polynomial $P(x) = x^4 - 3x^3 + ax^2 + bx - 6$ has a remainder of 8 when divided by $(x+1)$. Also, $(x-3)$ is a factor of $P(x)$. Find a and b .

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- (b) Consider the function $f(x) = 1 + \sqrt{x+1}$

- i) State the largest domain of $f(x)$ for which $f^{-1}(x)$ is defined.
- ii) Find $f^{-1}(x)$, stating its domain and range.
- ii) Sketch both $y = f(x)$ and $y = f^{-1}(x)$ on the same set of axes.
- iii) Find the point where $y = f(x)$ and $y = f^{-1}(x)$ intersect.

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Question 3 (12 marks)

- (a) Find the general solution of $\sqrt{2} \sin 2\theta + 1 = 0$ 2

- (b) i) Show that $e^{-x} - \cos x = 0$ has a root near $x = 1.31$.
ii) Use one step of Newton's method to find a closer approximation to the root. Answer to 3 decimal places. 4

- c) Joshua wishes to invest a regular amount ($\$M$) at the beginning of each quarter so that after five years he will have \$20 000. Interest is to be paid quarterly on the balance at 9% p.a.

- i) Show that after n quarters the balance of his investment is:

$$\frac{\$M(1+R)((1+R)^n - 1)}{R}$$

where R is the quarterly interest rate.

- ii) Hence or otherwise calculate M , the amount of each quarterly investment, to the nearest cent. 6

Question 4 (10 marks) Start a new page.

Marks:

- (a) Find all the real roots of $2x^3 - 3x^2 - 11x + 6 = 0$. 4
- (b) Rhys borrows \$94 000 to purchase his house. The loan is over 25 years at 10% p.a. interest, compounded on the balance owing before each monthly repayment. He used the following formula to calculate the value of his monthly repayments:

$$M = \frac{P(1+R)^n R}{((1+R)^n - 1)}$$

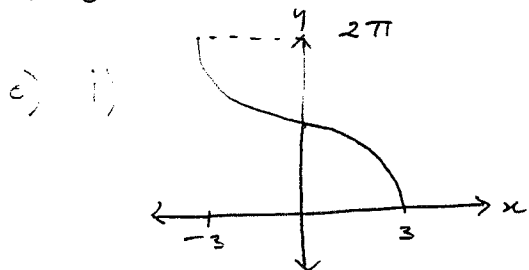
where P is the amount of the loan,
 R is the monthly interest rate,
 n is the number of months.

- i) Find the value of his monthly repayments, to the nearest cent.
- ii) If Rhys decides to pay \$1092 per month off his loan instead, how many years and months does he save on his loan?

Question 1.

a) $\frac{d}{dx} \tan^{-1} \frac{x}{2} = \frac{1}{1+(\frac{x}{2})^2} \cdot \frac{1}{2} = \frac{2}{4+x^2}$ 3U AS 2002 PLC

b) $\int \frac{1}{\sqrt{2-x^2}} dx = \sin^{-1} \frac{x}{\sqrt{2}} + c$

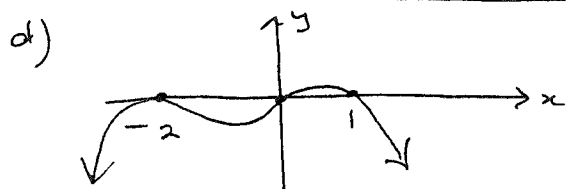


ii) Domain: $-1 \leq \frac{x}{3} \leq 1$

$\therefore -3 \leq x \leq 3$

Range $0 \leq \cos^{-1} \frac{x}{3} \leq \pi$

$\therefore 0 \leq 2\cos^{-1} \frac{x}{3} \leq 2\pi$



e) $\frac{du}{dx} = 2x$

When $x=1$ $u=0$

$x=2$ $u=3$

$\therefore \int_0^3 u^4 du = \left[\frac{u^5}{5} \right]_0^3 = \frac{243}{5}$

Question 2.

a) $P(-1) = 1 + 3 + a - b - 6 = 8$ (1)

$P(3) = 81 - 81 + 9a + 3b - 6 = 0$ (2)

(1) $a - b = 10$ $\therefore 4a = 12$

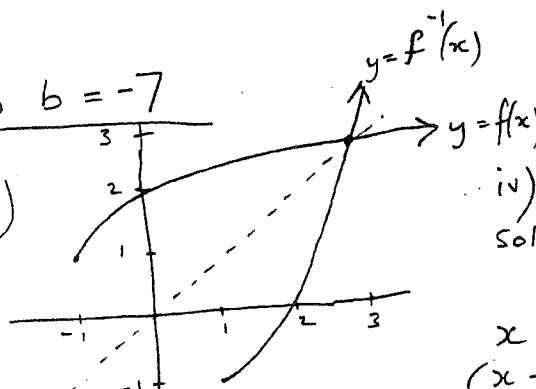
(2) $3a + b = 2$ $\Rightarrow a = 3, b = -7$

b) i) Domain $x \geq -1$

ii) $x = 1 + \sqrt{y+1}$

$(x-1)^2 = y+1$

iii)



iv) (3, 3)

Solve $y = x$
 $y = 1 + \sqrt{x+1}$

$x = 1 + \sqrt{x+1}$
 $(x-1)^2 = x+1$

Question 3.

a) $\sqrt{2} \sin 2\theta + 1 = 0$

$$\sin 2\theta = -\frac{1}{\sqrt{2}}$$

$$2\theta = n\pi + (-1)^n \sin^{-1}(-\frac{1}{\sqrt{2}})$$

$$\theta = \frac{n\pi}{2} + (-1)^{n+1} \frac{\pi}{8}$$

b) i) $f(1.29) = e^{-1.29} - \cos 1.29 = -0.0019$

$$f(1.32) = e^{-1.32} - \cos 1.32 = 0.019$$

Change in sign $\therefore$ root between.

ii) $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ $f(x) = e^{-x} - \cos x$
 $f'(x) = -e^{-x} + \sin x$

$$x_2 = 1.31 - \frac{e^{-1.31} - \cos 1.31}{(-e^{-1.31} + \sin 1.31)}$$

$$= 1.293$$

c) Let A_n = amount of investment after n quarters.

$$R = \frac{9\%}{4} = 2.25\%$$

$$A_1 = \$M(1+R)$$

$$A_2 = \$M(1+R)(1+R) + \$M(1+R)$$

$$A_n = \$M(1+R)^n + \$M(1+R)^{n-1} + \dots + \$M(1+R)$$

$$= \frac{\$M(1+R)[(1+R)^n - 1]}{1+R - 1} \quad \text{as required}$$

ii) After $5 \times 4 = 20$ quarters have \$20,000

$$\therefore \$20,000 = \frac{\$M(1.0225)(1.0225^{20} - 1)}{0.0225}$$

$$\$M = \$785.17$$

a) Let $P(x) = 2x^3 - 3x^2 - 11x + 6$

$$P(3) = 0 \therefore x-3 \text{ is a factor}$$

$$\begin{array}{r} (x-3) \overline{) 2x^3 - 3x^2 - 11x + 6} \\ \underline{2x^3 - 6x^2} \\ 3x^2 - 11x \\ \underline{3x^2 - 9x} \\ -2x + 6 \\ \underline{-2x + 6} \\ 0 \end{array}$$

$$\therefore P(x) = (x-3)(2x^2 + 3x - 2)$$

$$= (x-3)(2x-1)(x+2)$$

$$= 0 \text{ when } x = -2, \frac{1}{2}, 3$$

b) i) $M = \frac{94000 \left(1 + \frac{5}{600}\right)^{300} \cdot \frac{5}{600}}{\left(1 + \frac{5}{600}\right)^{300} - 1} = \854.18

$$R = \frac{10}{12}\% = \frac{5}{600}$$

$$n = 25 \times 12 = 300$$

months

ii) Let $M = \$1092$ $\$1092 = \frac{94000 \left(1 + \frac{5}{600}\right)^n \cdot \frac{5}{600}}{\left(1 + \frac{5}{600}\right)^n - 1}$

$$\Rightarrow 1092 \times \left(1 + \frac{5}{600}\right)^n - 1092 = 94000 \left(1 + \frac{5}{600}\right)^n \cdot \frac{5}{600}$$

$$\Rightarrow (1092 - 94000 \times \frac{5}{600}) \left(1 + \frac{5}{600}\right)^n = 1092$$

$$\Rightarrow n \ln \left(1 + \frac{5}{600}\right) = \ln \left(\frac{1092}{1092 - \frac{2350}{3}}\right)$$

$$n = \frac{1.2635}{0.00839} =$$

$$152.25$$

$$= 12 \text{ yrs } 9 \text{ mths}$$

$$= 12 \text{ yrs } 3 \text{ months}$$

$\therefore$ He saved

12 yrs 3 months